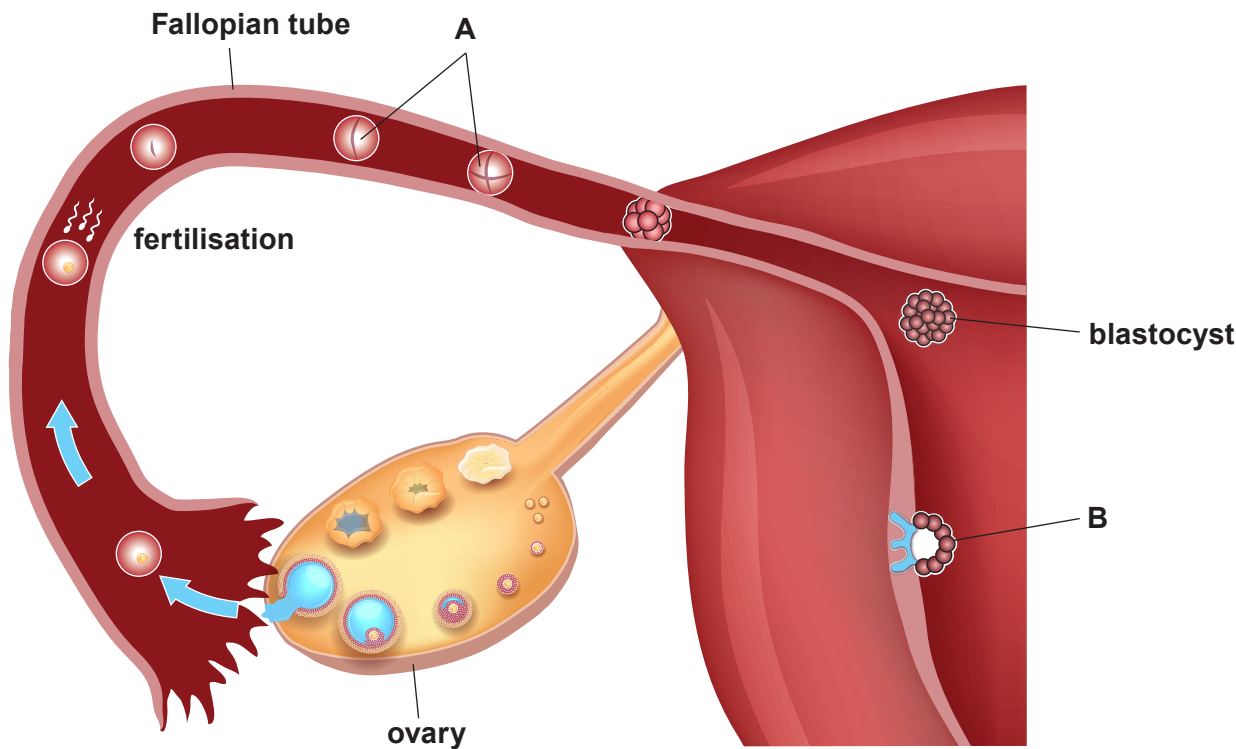


Answer **all** questions.

1. **Image 1.1** shows the sequence of events which occur in the fallopian tube and uterus before and after fertilisation.

Image 1.1



- (a) (i) State the process which has recently occurred in the ovary in **Image 1.1**. [1]
Process

- (ii) Briefly describe the process of fertilisation in humans. [4]

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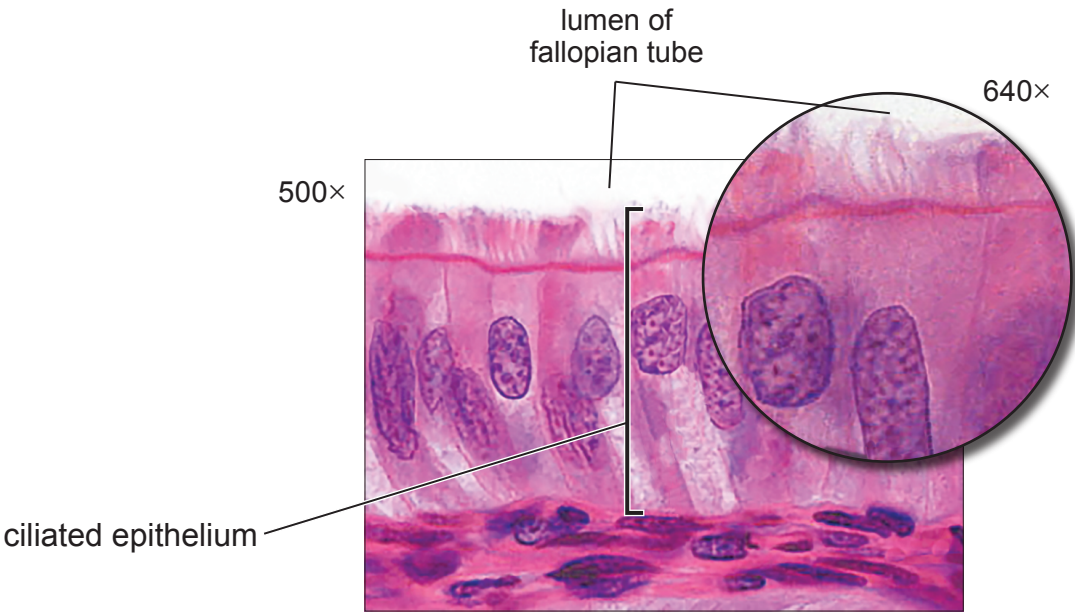
(iii) State the type of cell division involved in the production of structure **B** and the term given to the events shown at **A**. [1]

Type of cell division

Term given to events shown at **A**

Image 1.2 shows ciliated epithelium, which is found lining the fallopian tube, as seen with a light microscope.

Image 1.2



(b) Explain why ciliated epithelium is important in the sequence of events shown in **Image 1.1**. [3]

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- (c) In mammals, other than humans, the menstrual cycle is known as the oestrous cycle. During the oestrous cycle most female mammals are receptive to mating for only a limited period of time. This varies between mammals. The timing of ovulation also varies.

Table 1.3 shows information about the oestrous cycle of three different mammals.

Table 1.3

Average Reproductive Cycles				
Mammal	Length of oestrous cycle	Length of time receptive to male (oestrus)	Ovulation	(approx) Length of Pregnancy
cow	21 days (all year)	18 hr	11 hr after end of oestrus	9 months
sheep	17 days seasonal (autumn)	29 hr	near end of oestrus	5 months
horse	21 days seasonal (spring)	4–8 days	1–2 days before end of oestrus	12 months

- (i) Using the data shown in **Table 1.3**, explain why having a seasonal oestrous cycle is the best way to ensure offspring survival for animals such as sheep and horses. [2]

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- (ii) Explain why ovulation in the three species shown in **Table 1.3** occurs towards the end of oestrus (when the female is receptive to the male). [2]

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